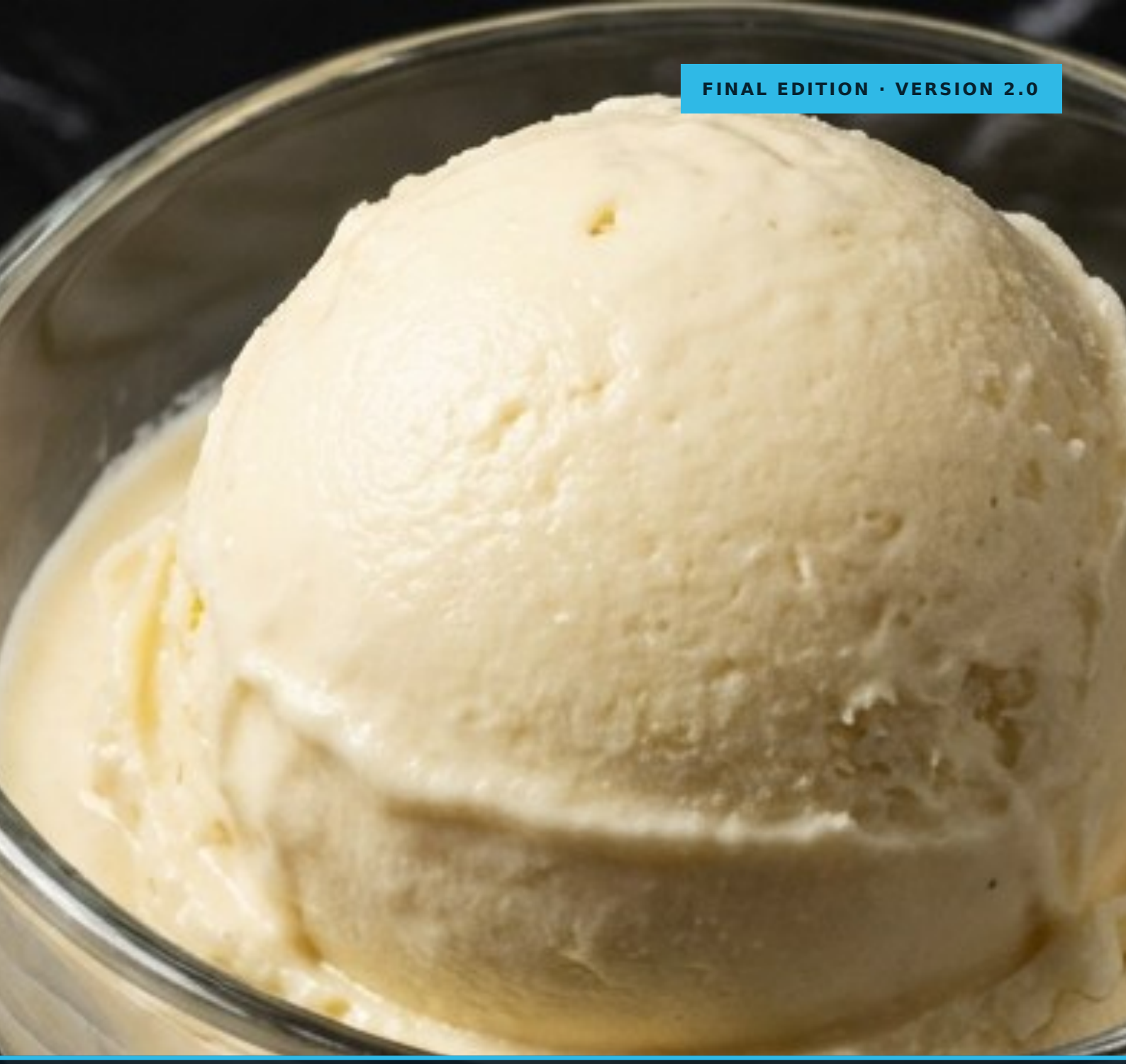


FINAL EDITION · VERSION 2.0

PROJECT 100 — KITCHEN LABORATORY

NINJA CREAMI XL PROTEIN ICE CREAM GUIDEBOOK

The synergized final edition: the complete operating system for high-protein, zero-sugar frozen desserts — universal base, 7-step workflow, 18 permutations, forensic validation — with every quantity in dual measure: kitchen units and true weight/volume.

§ ABOUT THIS FINAL EDITION

This Final Edition synergizes every edition of Bogdan’s Ninja Creami XL documentation into one print volume: the Master Operations Manual (Culinary Engine V198.4), the Master Architecture guidebook (Document Architecture V186.00) with its food-science layer, the earlier Operations Manual V194.1, and the V1.0 consolidated edition — including its validation pass against published references (SharkNinja’s NC501 Owner’s Guide, hydrocolloid manufacturer monographs, Health Canada / USDA guidance).

What changed in this edition. (1) **Dual measurement throughout:** every quantity in the book is printed in kitchen measure (scoops, tsp, tbsp, mL) and true measure (grams, milligrams, ounces, fluid ounces) — Chapter 3 is the system’s measurement standard. (2) The Master Architecture’s engineering layer is integrated: the 600–645 g fill-weight target, the allulose freezing-point science, the 0.25% anti-slime law, the three-tier inclusion architecture, and the pudding reduction directive. (3) All conflicts between editions are ruled on once, in Chapter 11 — nothing is silently overwritten, and the one procedure that runs against the manufacturer’s printed instruction (lid-off freezing) is labelled owner canon wherever it appears.

OPERATOR & SYSTEM PARAMETERS
Operator: **Bogdan Dospinescu**, age 46 · Chassis mass: 200.1 lb (90.8 kg) · Terminal target: 100.0 lb SMM (+11.8 lb)
Hardware: Ninja Creami XL Deluxe, NC501 series · 24 oz (709 mL) Deluxe Pints
System ceilings (PROJECT 100 planning architecture): 2,780 kcal/day · 252 g protein/day · **69 g lipid/day** (gallbladder cap — the reason the lipid tiers in Chapter 6 exist)
Pint target: ~80 g protein at zero sugar, per 24 oz (709 mL) pint

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CHAPTER 1 — THE SYSTEM AT A GLANCE

The Ninja Creami XL Deluxe (NC501 series) is not a churner and not a blender. In the Master Architecture's terms it is a precision vertical lathe: an 800-watt motor drives a dual-stage paddle down into a completely frozen solid block at roughly 1,800 RPM, micro-shaving microscopically thin layers off the frozen cylinder. Conventional ice cream leans on 14–20% butterfat and 15–22% sucrose to stay soft; a zero-sugar, 80 g-protein block instead freezes like a crystalline polymer — which is why surface geometry, stabilizer load and freeze discipline are hardware-protection issues, not culinary preferences.

The method is a 7-step pipeline in three phases, executed in strict order. Total fluid per pint: **600–645 g (21.2–22.8 oz)** in a 24 oz (709 mL) tub, never past the Max Fill line.

PHASE I · STEP 1

Prep

Dry matrix weighed to 0.1 g (0.004 oz); stabilizer calibrated; dry-whisked in isolation.

PHASE I · STEP 2

Emulsify

Hot-to-cold vortex: broth + collagen dissolved boiling, cold milk + egg whites, dry matrix streamed in over 90 seconds.

PHASE I · STEP 3

Settle

15-minute rest. Micro-foam dissipates. Fill weight lands at 600–645 g (21.2–22.8 oz).

PHASE II · STEP 4

Freeze

Lid-off 2–4 h, then lid on, 24 h total at −18 °C to −20 °C (0 °F to −4 °F). Owner canon — see Ch. 10.

PHASE II · STEP 5

Level

Solid-block check, Deiss masher levelling, 30–60 s warm perimeter bath.

PHASE III · STEP 6

Spin

Full zone, Lite Ice Cream (Sorbet for isolate bases). Wall scrape, tuned re-spin, inclusions by tier.

PHASE III · STEP 7

Flush

Leftovers pressed flat; blade and spindle flushed with hot water immediately. Critical, not housekeeping.



THE MACHINE — Ninja Creami XL (NC501 series): Ice Cream, Lite Ice Cream, Sorbet, Gelato, Frozen Yogurt, Creamiccino, Frozen Drink, Slushi, Milkshake, Italian Ice — plus Mix-In and Re-Spin. Owner photo.

CHAPTER 2 — THE PHYSICS & FOOD SCIENCE OF THE BASE

Hydrophobicity and stabilization

Protein isolates resist water integration. Dry protein or gum dropped straight into liquid forms hydrophobic “glue pockets” that freeze into icy lumps and load the paddle with dangerous resistance. Hence the system’s first law: stabilizer is always isolated inside the dry matrix before liquid appears (Step 1), then streamed into a moving vortex (Step 2). Gum manufacturers prescribe the identical technique — pre-blend with dry powders to prevent “fish-eye” clumps (Chapter 10, Claim 7).

Volumetric expansion and the centre hump

Egg-white albumin and whey trap micro-air during blending and give the base its overrun — but frozen under a sealed lid, that expansion drives the centre upward into a frost hump. A convex surface deflects the descending spindle and loads the drive with lateral shear. The countermeasures are layered: the 15-minute settle, the lid-off first freeze phase, and masher levelling before any spin. Ninja’s manual forbids processing a pint frozen unevenly or at an angle; the hump itself is documented in community forensic practice (Chapter 10, Claim 2).

Freezing-point depression — the allulose engine

Freezing-point depression is colligative: it scales with the number of dissolved particles, not their mass. Allulose (molecular weight 180.16 g/mol) delivers nearly twice the particle count per gram of sucrose (342.3 g/mol) — a freezing-point depression factor of **1.9×**. The system’s dose of **25–35 g (0.9–1.2 oz)** of crystalline allulose per pint therefore softens the frozen matrix measurably, without the crystallization-in-storage that hardens erythritol-sweetened pints, and without sugar. Published references confirm allulose depresses the freezing point and supports scoopability (Chapter 10, Claim 9), with the nuance recorded there.

The hydrocolloid anti-slime law

Xanthan, guar and tara are pseudoplastic, shear-thinning hydrocolloids. In a plain isolate base they suppress ice crystals cleanly at 1/4 tsp (1.0 g · 1000 mg). But in matrices that already carry whey/casein blends and modified starches (instant pudding mixes), total hydrocolloid load above roughly **0.25% w/w** cross-links into an elastic, rubbery slime. That threshold is the physical basis of the calibration rule in Chapter 4: blends and pudding bases are halved to 1/8 tsp (0.5 g · 500 mg).

The taste-audit law

Sub-zero temperatures temporarily anesthetize taste receptors: cold dulls sweetness and flavour. The warm base must therefore taste noticeably over-sweet at the pre-freeze audit; a base that tastes merely mild warm will eat flat and unpalatable frozen. This is standard ice-cream doctrine, confirmed in the literature (Chapter 10, Claim 15).

What the collagen is — and is not

Gelatin is a classic ice-cream stabilizer. Hydrolyzed collagen peptides are not gelatin: per GELITA’s technical data they cannot form a gel network at any useful concentration. In this system, bone broth

and marine collagen are a **protein payload** — amino acids and macro mass. Stabilization belongs to the calibrated gums alone (Chapter 10, Claim 10).

CHAPTER 3 — THE MEASUREMENT STANDARD

Every quantity in this book is dual-printed: the kitchen measure the hand uses, and the true measure the scale verifies. This chapter is the conversion authority for the whole system.

Master conversions

Unit	Metric	Imperial	Notes
1 tsp	5 mL	—	Gum doses: 1/4 tsp = 1.0 g; 1/8 tsp = 0.5 g (levelled flush, Ch. 4)
1 tbsp	15 mL · 3 tsp	0.5 fl oz	Allulose: 1 tbsp ≈ 8–9 g · Black cocoa: 1 tbsp ≈ 5 g
1 fl oz	29.57 mL	—	Re-spin splash: 15–30 mL = 0.5–1.0 fl oz = 1–2 tbsp
1 oz (weight)	28.35 g	—	Pint fill: 600–645 g = 21.2–22.8 oz
1 mg	0.001 g	—	Gum in mg: 1/4 tsp = 1000 mg · 1/8 tsp = 500 mg
24 oz pint	709 mL	3 cups	Max Fill line sits below the lid threads
68 oz pitcher	2.0 L	—	The 3× batch vessel (Ch. 4)
Temperature	−18 °C · −20 °C · −25 °C	0 °F · −4 °F · −13 °F	Working freeze window · working window · cement-brick floor

Scoop & ingredient equivalents (this system's products)

Item	Kitchen measure	True measure
Target protein powder (Hexapro / PVL Isogold / Syntha-6)	1.5 scoops	~38–45 g (1.3–1.6 oz)
Bone broth powder (Left Coast Performance)	1 scoop	~10 g (0.35 oz)
Marine collagen (Organika / pure)	1 scoop	~10 g (0.35 oz)
Granulated allulose / sweetener core	3–4 tbsp	25–35 g (0.9–1.2 oz)
Stabilizer gum — isolate bases	1/4 tsp, levelled	1.0 g (1000 mg)
Stabilizer gum — blend / pudding bases	1/8 tsp, levelled	0.5 g (500 mg)
Kirkland almond butter — Tier 1	1 tsp	5 g (0.18 oz)
Kirkland almond butter — Tier 2	1 tbsp	15 g (0.53 oz)
Cacao nibs / frozen SF mini chips	~2 tsp	10 g (0.35 oz)

Defatted peanut powder	2 tbsp / 3 tbsp	~12 g / ~18 g (0.42 / 0.63 oz)
Black cocoa powder	1 tbsp	~5 g (0.18 oz)
Liquid egg whites, pasteurized	~1/3 cup + 1 tbsp	100 g (3.5 oz) ≈ 100 mL
Kirkland ultra-filtered 2% milk	~1 cup + 1 tbsp	250 mL (8.5 fl oz)
Filtered water, boiling (70–100 °C / 158–212 °F)	—	150–200 mL (5.1–6.8 fl oz)
LorAnn emulsions / vanilla paste	1/2–1 tsp	2.5–5 mL
Pink Himalayan salt	1 generous pinch	~0.5 g (500 mg)

Spoon-to-weight figures are for the specific products named, levelled with the flat bar — a mounded spoon of gum can double the dose, which is precisely the slime trap (Chapter 12). When a kitchen measure and the scale disagree, the scale governs: this system weighs to 0.1 g (0.004 oz) on a 2000 g (4.4 lb) capacity scale.

CHAPTER 4 — THE UNIVERSAL STRUCTURAL BASE FORMULA

System Architecture V198.4. One base, held static across all 18 permutations, so fluid-to-solid ratio, ice-crystal suppression and protein hydration never drift. Every quantity is dual-printed per the Chapter 3 standard. **Finished fill weight: 600–645 g (21.2–22.8 oz)** per 24 oz (709 mL) pint — if the scale lands outside that window, the base is wrong before it is frozen.

1 · Thermal phase — the high-hydration start

Ingredient	Kitchen measure	True measure	Function
Filtered water, boiling (70–100 °C / 158–212 °F)	—	150–200 mL (5.1–6.8 fl oz)	High-temperature hydration
Left Coast Performance Bone Broth Powder	1 scoop	~10 g (0.35 oz)	Protein payload; amino acids
Pure Marine Collagen (Organika)	1 scoop	~10 g (0.35 oz)	Protein payload (not a stabilizer — Ch. 2)

2 · Cold phase — the universal density matrix

Ingredient	Kitchen measure	True measure	Yield
Kirkland Signature Ultra-Filtered 2% Milk	~1 cup + 1 tbsp	250 mL (8.5 fl oz)	13 g protein · 5.0 g butterfat · 6 g net carbs · 0 g lactose
Liquid egg whites, pasteurized	~1/3 cup + 1 tbsp	100 g (3.5 oz) ≈ 100 mL	Albumin overrun; structure

3 · Dry integration matrix — stabilizer-calibrated

Ingredient	Kitchen measure	True measure	Notes
Target protein powder (Hexapro / PVL Isogold / Syntha-6)	1.5 scoops	~38–45 g (1.3–1.6 oz)	30–38 g high-BV protein by brand
Zero-calorie sweetener core (allulose / monk fruit / brown)	3–4 tbsp	25–35 g (0.9–1.2 oz)	Allulose portion drives freezing-point depression (Ch. 2)
Stabilizer — standard isolate bases (PVL Isogold, no pudding)	1/4 tsp, levelled	1.0 g (1000 mg)	Tara, xanthan, or guar
	1/8 tsp, levelled	0.5 g (500 mg)	Halved: the anti-slime law (Ch. 2)

Stabilizer — thick blends (Hexapro / Syntha-6) or any pudding base			
Pink Himalayan salt	1 generous pinch	~0.5 g (500 mg)	Flavour sharpener; assists freezing-point depression
Flavour extracts / pastes (LorAnn, Vanillux)	1/2–1 tsp	2.5–5 mL	Per permutation (Ch. 8). Paste carrier must be glycerin or alcohol — never agave or glucose syrup

> **THE PUDDING REDUCTION DIRECTIVE**
> Sugar-free instant pudding mixes are stabilizer systems in themselves — modified corn starch plus tetrasodium pyrophosphate. Any permutation carrying 1 tbsp (15 mL · ~8 g) of pudding mix drops to the 1/8 tsp (0.5 g · 500 mg) gum line automatically, regardless of base powder. Applies to Permutations 17 and 18.

The Silk variant (V194.1 base)

The earlier architecture ran the identical structure on **Silk Protein Almond & Cashew Milk, 200–250 mL (6.8–8.5 fl oz)**, in place of the Kirkland milk. It remains the certified dairy-free variant. Its macro accounting differs (Chapter 9 prints both ledgers); on Permutations 01–07 the Silk pint runs 464 kcal · 79.5 g protein · 11 g fat · 13 g net carbs — the numbers on the printed kitchen cards.

Batch scaling — the 68 oz (2.0 L) pitcher protocol, 3× pints

Component	Single pint	3× pitcher
Boiling filtered water	150–200 mL (5.1–6.8 fl oz)	450–600 mL (15.2–20.3 fl oz)
Bone broth + collagen	1 scoop each (~10 g / 0.35 oz each)	3 scoops each (~30 g / 1.1 oz each)
Milk (Kirkland 2% or Silk variant)	250 mL (8.5 fl oz)	750 mL (25.4 fl oz)
Liquid egg whites	100 g (3.5 oz)	300 g (10.6 oz)
Protein powder	1.5 scoops (~38–45 g / 1.3–1.6 oz)	4.5 scoops (~115–135 g / 4.1–4.8 oz)
Sweetener core	3–4 tbsp (25–35 g / 0.9–1.2 oz)	9–12 tbsp (75–105 g / 2.6–3.7 oz)
Stabilizer gum	1/4 or 1/8 tsp (1.0 or 0.5 g)	3/4 or 3/8 tsp (3.0 or 1.5 g · 3000 or 1500 mg)
Pink salt	1 pinch (~0.5 g)	3 pinches (~1.5 g)

Bulk workflow mirrors the single-pint pipeline: dry components whisked together; broth and collagen dissolved in the boiling water in the pitcher; cold milk and egg whites added; vortex established; dry matrix streamed in over 90 seconds; blend high 30 s, rest 2 min for stabilizer hydration, final 15 s burst; 15-minute settle; portion into three 24 oz (709 mL) tubs at 600–645 g (21.2–22.8 oz) each.

CHAPTER 5 — THE 7-STEP WORKFLOW

Step 1 · Precision dry calibration

Dry bowl on the digital scale — 2000 g (4.4 lb) capacity, 0.1 g (0.004 oz) readability. Weigh the protein powder (1.5 scoops · ~38–45 g / 1.3–1.6 oz), sweetener (3–4 tbsp · 25–35 g / 0.9–1.2 oz), salt (1 pinch · ~0.5 g) and any spice enhancers. Measure the gum with the magnetic spoon scraped perfectly flush with the flat metal levelling bar — 1/4 tsp (1.0 g · 1000 mg) for isolates, 1/8 tsp (0.5 g · 500 mg) for blends and pudding bases. Dry-whisk thoroughly so gum particles are isolated among dry powder before any liquid exists.

Step 2 · Thermal emulsion vortex

Dissolve the bone broth (1 scoop · ~10 g) and collagen (1 scoop · ~10 g) into the boiling water (150–200 mL · 5.1–6.8 fl oz, 70–100 °C / 158–212 °F) until translucent. If the permutation carries almond butter, it dissolves in this hot phase — heat melts the lipids for a clean micro-emulsion. Temper with the cold milk (250 mL · 8.5 fl oz) and egg whites (100 g · 3.5 oz). Submerge the KitchenAid immersion blender, establish a low-speed vortex, and stream the dry matrix in over a continuous 90 seconds.

Pre-freeze taste audit (critical): the warm base must taste noticeably over-sweet (Chapter 2). If it tastes mild, add 1 tbsp (15 mL · ~8 g) sweetener now. There is no correction after the freeze.

Step 3 · The 15-minute micro-foam settle

Rest the base undisturbed for exactly 15 minutes; whipped micro-air rises and collapses. Pour into a clean 24 oz (709 mL) tub below the Max Fill line, tap the base firmly on the counter three times to evacuate deep bubbles, and verify: **600–645 g (21.2–22.8 oz)** on the scale.

Step 4 · Stratified lid-off freezing — owner canon

Freeze the tub uncovered for the first 2–4 hours at −18 °C to −20 °C (0 °F to −4 °F): unrestricted vertical expansion prevents the centre dome from ever forming. Snap the lid on and freeze undisturbed to a total of 24 hours (16 hours absolute minimum) at the same window. Avoid deep chest freezers below −25 °C (−13 °F) — cement-brick territory that even Ninja's manual warns against.

> **DOCUMENTED DIVERGENCE — OWNER CANON**
> Ninja's manual instructs freezing with the lid snapped on for 24 hours. The lid-off first phase is established community technique (Tasty Travelers, Kroll's Korner) adopted as owner canon in this system because it prevents the hump at the source. It is labelled wherever it appears (Chapter 10, Claim 3) and never presented as manufacturer guidance.

Step 5 · Masher levelling & perimeter bath

1. **Solid-block check:** the pint must be 100% frozen solid. Any soft centre → back to the freezer for 4 hours.
2. **Perimeter bath:** warm tap water over the lower half of the tub's outer walls for 30–60 seconds — never past 60 s.
3. **Levelling:** press the Deiss stainless masher straight down, firmly, shattering any frost peak into a perfectly horizontal plane. Never twist — tub walls crack.

Step 6 · Mechanical processing & re-spin tuning

Lock the tub into the outer bowl, selector to **Full**, program **Lite Ice Cream** — or **Sorbet** for pure isolate bases, whose higher shear duration suits dense isolate matrices. Never standard Ice Cream on these bases (Chapter 10, Claim 4). After the first spin:

- **Wall scrape:** run a butter knife gently around the inner wall — the paddle leaves a ~1 mm (0.04 in) frost ring it cannot reach — and fold those crystals into the core. Gentle pressure; the manual cautions against aggressive metal utensils in the pint.
- **Dry re-spin first:** for hard-scoop gelato density, re-spin with no added liquid; blade friction binds the mass. This is Ninja's own NC501 procedure (Claim 5).
- **Splash re-spin:** only if the pint stays powdery — 15–30 mL (1–2 tbsp · 0.5–1.0 fl oz) of cold milk into the centre cavity, then re-spin, for soft-serve texture. Liquid added prematurely makes soup.
- **Inclusions:** strictly by the three-tier architecture, Chapter 7.

Step 7 · Post-spin refreeze & spindle flush — critical

Leftovers are scraped and pressed 100% flat with a metal spoon before refreezing — a peak refrozen is a spindle hazard at the next spin, and Ninja's manual requires the same flattening. Then, immediately: separate the blade assembly from the lid and flush hot running water directly through the central spindle aperture. Aerosolized dairy and protein mist migrate into the drive-shaft recess on every spin; unwashed assemblies grow biofilm and seize their locking springs. This step protects the machine, not the counter.

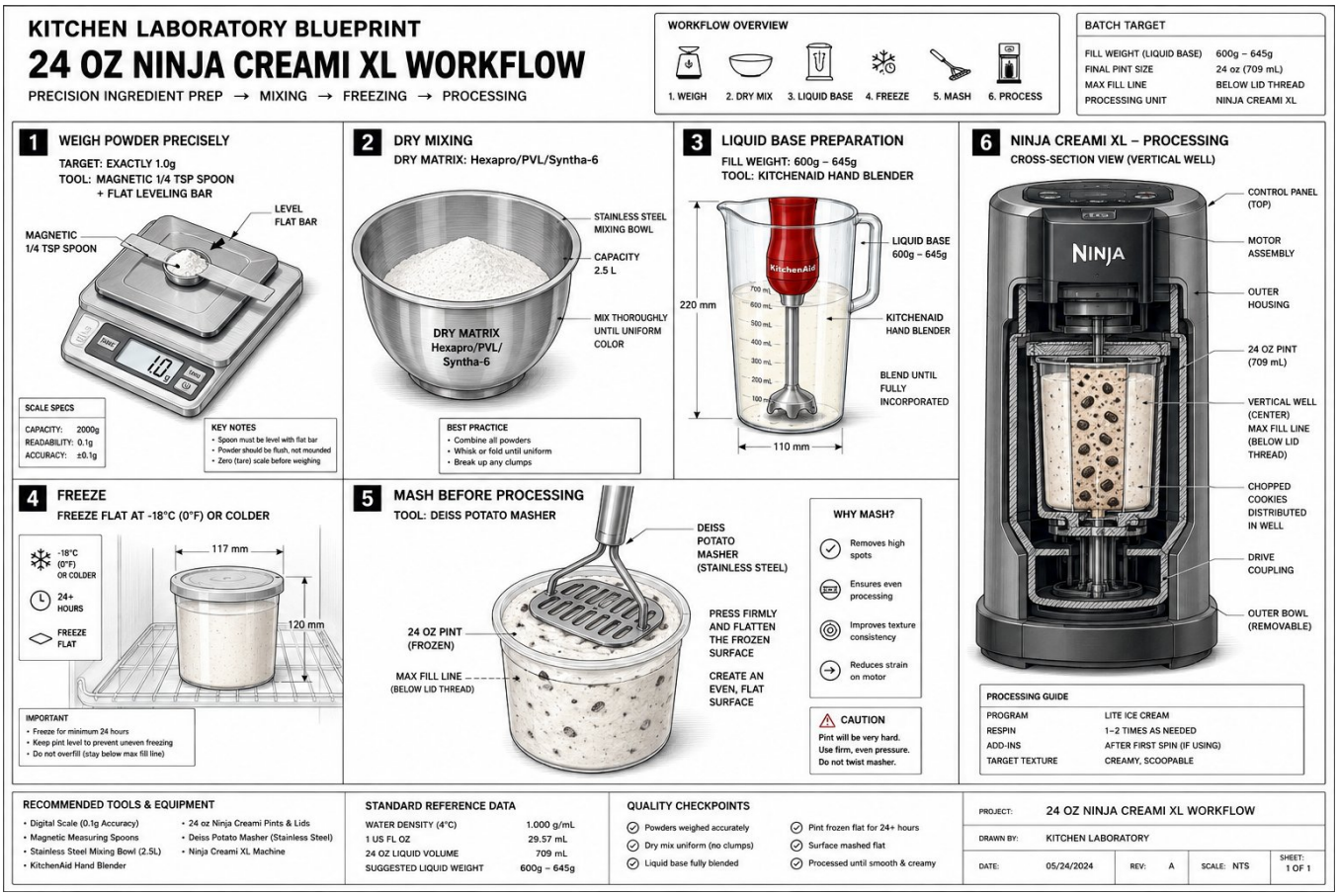


PLATE — Kitchen Laboratory blueprint: weigh → dry mix → liquid base → freeze → mash → process, with the 600–645 g fill-weight specification. Owner artwork.

CHAPTER 6 — THE LIPID & FAT MATRIX

The Kirkland 2% rule

The base milk is the fat system: 250 mL (8.5 fl oz) of Kirkland ultra-filtered 2% supplies **5.0 g of natural, emulsified dairy butterfat** — native ice-crystal suppression that arrives pre-emulsified — with 13 g protein, 6 g net carbs and 0 g lactose. Added nut fats sit under two regulated tiers, calibrated against the operator's **69 g daily lipid ceiling**: fat is budgeted, never free-poured.

Tier	Dose (dual)	Macro impact	Target use
Base alone	250 mL (8.5 fl oz) Kirkland 2%	5.0 g fat · 13 g protein · 6 g net carbs	Every pint
Tier 1 — micro-dose	1 tsp (5 mL) almond butter = 5 g (0.18 oz)	+2.7 g fat · +1.3 g protein · +33 kcal	Mouthfeel lift mirroring French vanilla custard (Permutation 04)
Tier 2 — max-dose	1 tbsp (15 mL) almond butter = 15 g (0.53 oz)	+8.0 g fat · +4.0 g protein · +100 kcal	Authentic nut profile (Permutations 11, 14, 15)

Hot emulsion SOP: any almond butter dose enters during the boiling-water phase of Step 2, before the cold milk. High heat melts the lipids so the immersion blender forms a seamless emulsion — cold-added nut butter freezes as waxy flecks.

CHAPTER 7 — THE THREE-TIER INCLUSION ARCHITECTURE

Inclusions are classified by how they survive the paddle. The class decides the method — there are no exceptions.

Tier 1 · Machine borehole — approved

Members: cacao nibs (10 g · 0.35 oz), frozen zero-sugar chocolate chips and mini chips. **Protocol:** inclusions must be frozen solid in advance. After the spin, drill a vertical borehole roughly 1 in (2.5 cm) wide to the bottom of the tub with a knife, pour the inclusions into the cavity — never piled on top — and run the dedicated **Mix-In** program. Ninja's manual sets the same architecture: a 1½-inch hole to the bottom, chopped mix-ins, ~1/4 cup (60 mL · ~35–40 g) maximum (Chapter 10, Claim 12).

Tier 2 · Manual hand-fold — never machine

Members: sugar-free protein dough bits, crunch nuggets, protein-bar chunks. **Protocol: never** the Mix-In program — high-velocity blades pulverize soft inclusions into muddy slime. Drop them into the spun pint and fold gently with a metal spoon.

Tier 3 · Manual knife-swirl — ribbons

Members: zero-calorie caramel syrup, sugar-free chocolate fudge drizzle. **Protocol:** carve 3–4 vertical channels through the processed pint with a butter knife, drizzle the syrup into the channels, and swirl gently twice for artisan marbling.



THE PANTRY — the emulsion library (LorAnn: Coffee, Vanilla, Lemon, Peppermint, Strawberry, Butter Vanilla, Blueberry; Vanillux vanilla paste), OrgunFun pasteurized egg white protein, Himalayan pink salt, Bob’s Red Mill xanthan gum, Creami XL tubs, Mio drops, VOLTRX shaker. Owner photo.

CHAPTER 8 — THE 18 ZERO-SUGAR PERMUTATIONS

Every permutation runs on the Chapter 4 base, unchanged. Flavour quantities are dual-printed (Chapter 3 standard). Macros are per full 24 oz (709 mL) pint on the Kirkland base ledger (Chapter 9).

01 · Lemon Chiffon

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) LorAnn Lemon Emulsion + 1/2 tsp (2.5 mL) SF Vanilla Paste

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar — lemon-curd tang

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

420 kcal · 78.5 g P · 7.5 g F · 9.5 g NC

02 · Strawberry Cream

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) LorAnn Strawberry Emulsion + 1/2 tsp (2.5 mL) SF Vanilla Paste

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar — berry lift

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

420 kcal · 78.5 g P · 7.5 g F · 9.5 g NC

03 · Blueberry Muffin

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) LorAnn Blueberry Emulsion

Enhancer: + 1/4 tsp (~0.65 g) Ceylon Cinnamon + 1/8 tsp (~0.4 g) Cream of Tartar

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

421 kcal · 78.5 g P · 7.5 g F · 9.7 g NC

04 · Classic Vanilla Bean

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) SF Vanilla Paste + 1 tsp (5 g · 0.18 oz) Almond Butter — Tier 1

Enhancer: Standard baseline

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

453 kcal · 79.8 g P · 10.2 g F · 10.2 g NC

05 · Velvet Banana Cream

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) LorAnn Banana Cream + 1/2 tsp (2.5 mL) LorAnn Butter Vanilla

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar — cream-pie tang

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

420 kcal · 78.5 g P · 7.5 g F · 9.5 g NC

06 · Spiced Chai Latte

Base: Hexapro Vanilla

Flavour core: 1/8 tsp (~0.25 g) Cardamom + 1 tsp (~2 g) Black Tea Powder

Enhancer: + 1/2 tsp (~1.3 g) Ceylon Cinnamon — chai spice core

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

422 kcal · 78.5 g P · 7.5 g F · 10.0 g NC

07 · Pistachio Biscotti

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) Pistachio Flavour + 1/4 tsp (1.25 mL) Premium Almond Extract

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar — biscotti dough tang

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

420 kcal · 78.5 g P · 7.5 g F · 9.5 g NC

08 · Midnight Mocha

Base: PVL Isogold (Iced Mocha)

Flavour core: 1 tsp (5 mL) Coffee Emulsion + 1/2 tsp (~1 g) Instant Café Bustelo

Enhancer: Standard baseline

Stabilizer: 1/4 tsp = 1.0 g (1000 mg) — pure isolate

380 kcal · 81.5 g P · 6.0 g F · 8.0 g NC

09 · Triple Mint Chip

Base: PVL Isogold (Triple Choc)

Flavour core: 1/2 tsp (2.5 mL) Peppermint Emulsion + 10 g (0.35 oz) Frozen Cacao Nibs — Tier 1 Mix-In

Enhancer: Standard baseline

Stabilizer: 1/4 tsp = 1.0 g (1000 mg) — pure isolate

430 kcal · 82.5 g P · 10.5 g F · 9.0 g NC

10 · Dark Chocolate Oreo

Base: PVL Isogold (Triple Choc)

Flavour core: 1 tbsp (~5 g) Black Cocoa + 1/2 tsp (2.5 mL) SF Vanilla Paste + 10 g (0.35 oz) Cacao Nibs

Enhancer: Standard baseline

Stabilizer: 1/4 tsp = 1.0 g (1000 mg) — pure isolate

450 kcal · 83.5 g P · 11.0 g F · 10.0 g NC

11 · Peanut Butter Twist

Base: PVL Isogold (Iced Mocha)

Flavour core: 2 tbsp (~12 g) Defatted PB Powder + 1 tbsp (15 g · 0.53 oz) Almond Butter — Tier 2

Enhancer: Standard baseline

Stabilizer: 1/4 tsp = 1.0 g (1000 mg)

510 kcal · 91.5 g P · 12.0 g F · 11.0 g NC

12 · Birthday Cake

Base: Syntha-6 (Vanilla)

Flavour core: 1 tsp (5 mL) Butter Vanilla + 1/2 tsp (2.5 mL) SF Vanilla Paste

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar — cake-batter tang

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

460 kcal · 74.0 g P · 11.0 g F · 21.0 g NC

13 · Cinnamon Roll Glaze / Snickerdoodle

Base: Syntha-6 (Vanilla)

Flavour core: 1 tsp (5 mL) Butter Vanilla + 1/2 tsp (2.5 mL) Maple Emulsion

Enhancer: + 1/2 tsp (~1.3 g) Ceylon Cinnamon + 1/8 tsp (~0.4 g) Cream of Tartar

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

462 kcal · 74.0 g P · 11.0 g F · 21.2 g NC

14 · Maple Walnut Cream

Base: Syntha-6 (Vanilla)

Flavour core: 1.5 tsp (7.5 mL) Maple Emulsion + 1 tbsp (15 g · 0.53 oz) Almond Butter — Tier 2

Enhancer: + 1/4 tsp (~0.65 g) Ceylon Cinnamon

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

561 kcal · 78.0 g P · 19.0 g F · 22.2 g NC

15 · PB Fudge Crunch

Base: Hexapro Vanilla

Flavour core: 3 tbsp (~18 g) Defatted PB Powder + 1 tbsp (15 g · 0.53 oz) Almond Butter — Tier 2 + 10 g (0.35 oz) Cacao Nibs

Enhancer: Standard baseline

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

610 kcal · 92.5 g P · 18.5 g F · 14.5 g NC

16 · Zero-Sugar Cookie Dough

Base: Hexapro Vanilla

Flavour core: 1 tsp (5 mL) SF Vanilla Paste + 1 tbsp (~8 g) Zero-Cal Brown Sweetener

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar + 1/8 tsp (~0.35 g) Ceylon Cinnamon

Inclusions: SF Protein Dough Bits — Tier 2 hand-fold

Stabilizer: 1/8 tsp = 0.5 g (500 mg)

471 kcal · 83.5 g P · 10.0 g F · 11.7 g NC

17 · Zero-Sugar Caramel Crunch

Base: Hexapro Vanilla

Flavour core: 1 tbsp (15 mL) Zero-Cal Caramel + 1 tbsp (~8 g) SF Butterscotch Pudding Mix

Enhancer: + 1/8 tsp (~0.4 g) Cream of Tartar + 1/8 tsp (~0.35 g) Ceylon Cinnamon

Inclusions: SF Protein Crunch Bits — Tier 2 hand-fold · zero-cal caramel — Tier 3 swirl

Stabilizer: 1/8 tsp = 0.5 g (500 mg) — pudding directive

461 kcal · 81.5 g P · 8.5 g F · 12.2 g NC

18 · Zero-Sugar Chocolate Swirl

Base: PVL Isogold (Triple Choc)

Flavour core: 1 tbsp (~5 g) Black Cocoa + 1 tbsp (~8 g) SF Chocolate Pudding Mix

Inclusions: 10 g (0.35 oz) Frozen SF Mini Chips — Tier 1 Mix-In · SF fudge — Tier 3 knife-swirl

Enhancer: Standard baseline

Stabilizer: 1/8 tsp = 0.5 g (500 mg) — pudding directive

440 kcal · 85.5 g P · 8.5 g F · 10.0 g NC

Culinary notes by line

Fruit & Bakery line (01–03): the 1/8 tsp (~0.4 g) of cream of tartar supplies the authentic acidic tang of lemon curd, berry preserves or buttermilk bakery batter — brightness without citric-acid crystal bite. Its role is flavour (Chapter 10, Claim 14). **Custard & Spice line (04–07):** Ceylon cinnamon gives coumarin-free warmth; the Tier 1 almond-butter micro-dose in Classic Vanilla Bean mirrors the lipid richness of French vanilla custard. **Isolate Performance line (08–09 and the PVL bases):** PVL Isogold carries no casein-micelle bulk, so the full 1/4 tsp (1.0 g · 1000 mg) stabilizer line is mandatory to prevent ice-crystal flaking and hold dense Italian gelato texture.

CHAPTER 9 — MACRO LEDGERS

Per full 24 oz (709 mL) pint. The **V198.4 ledger** (Kirkland 2% base) governs this edition. The **V194.1 ledger** (Silk base) is preserved beneath it — the grids differ because the base milk changed, not because a recipe did.

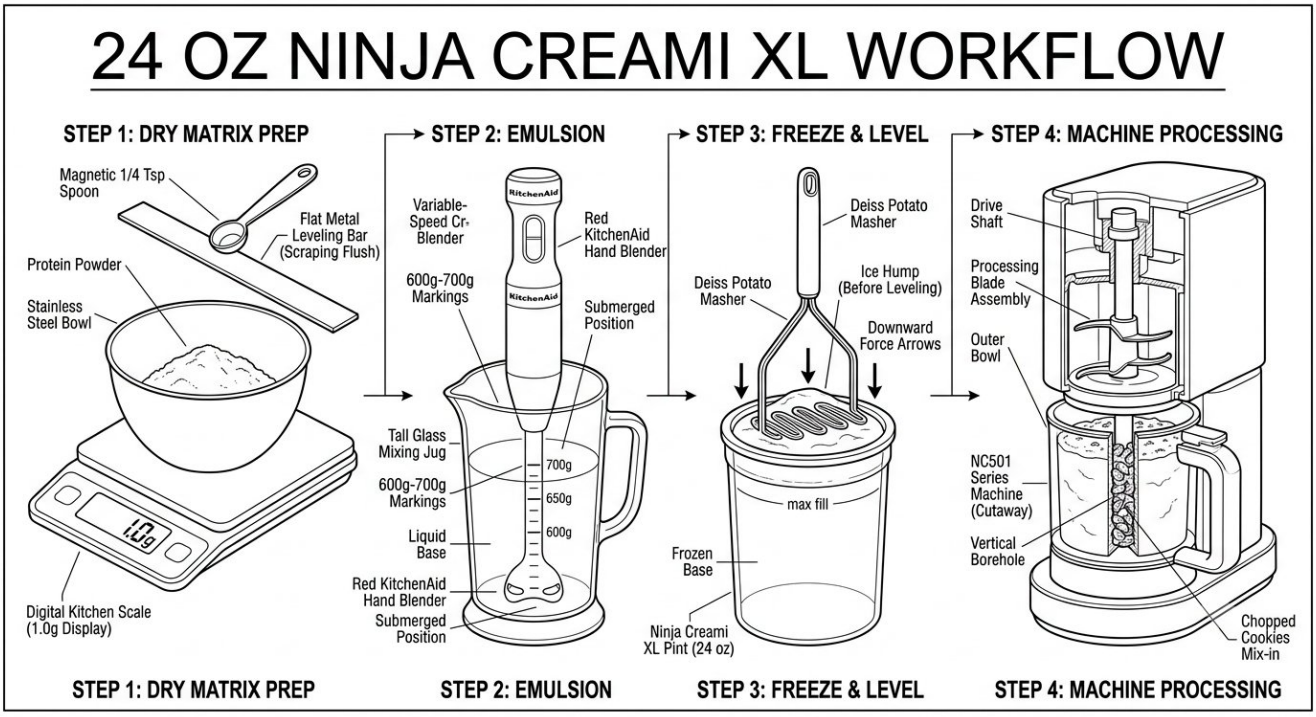
V198.4 — Kirkland base (governing)

#	Permutation	kcal	Protein	Fat	Net carbs
01	Lemon Chiffon	420	78.5 g	7.5 g	9.5 g
02	Strawberry Cream	420	78.5 g	7.5 g	9.5 g
03	Blueberry Muffin	421	78.5 g	7.5 g	9.7 g
04	Classic Vanilla Bean	453	79.8 g	10.2 g	10.2 g
05	Velvet Banana Cream	420	78.5 g	7.5 g	9.5 g
06	Spiced Chai Latte	422	78.5 g	7.5 g	10.0 g
07	Pistachio Biscotti	420	78.5 g	7.5 g	9.5 g
08	Midnight Mocha	380	81.5 g	6.0 g	8.0 g
09	Triple Mint Chip	430	82.5 g	10.5 g	9.0 g
10	Dark Chocolate Oreo	450	83.5 g	11.0 g	10.0 g
11	Peanut Butter Twist	510	91.5 g	12.0 g	11.0 g
12	Birthday Cake	460	74.0 g	11.0 g	21.0 g
13	Cinnamon Roll Glaze	462	74.0 g	11.0 g	21.2 g
14	Maple Walnut Cream	561	78.0 g	19.0 g	22.2 g
15	PB Fudge Crunch	610	92.5 g	18.5 g	14.5 g
16	Zero-Sugar Cookie Dough	471	83.5 g	10.0 g	11.7 g
17	Zero-Sugar Caramel Crunch	461	81.5 g	8.5 g	12.2 g
18	Zero-Sugar Chocolate Swirl	440	85.5 g	8.5 g	10.0 g

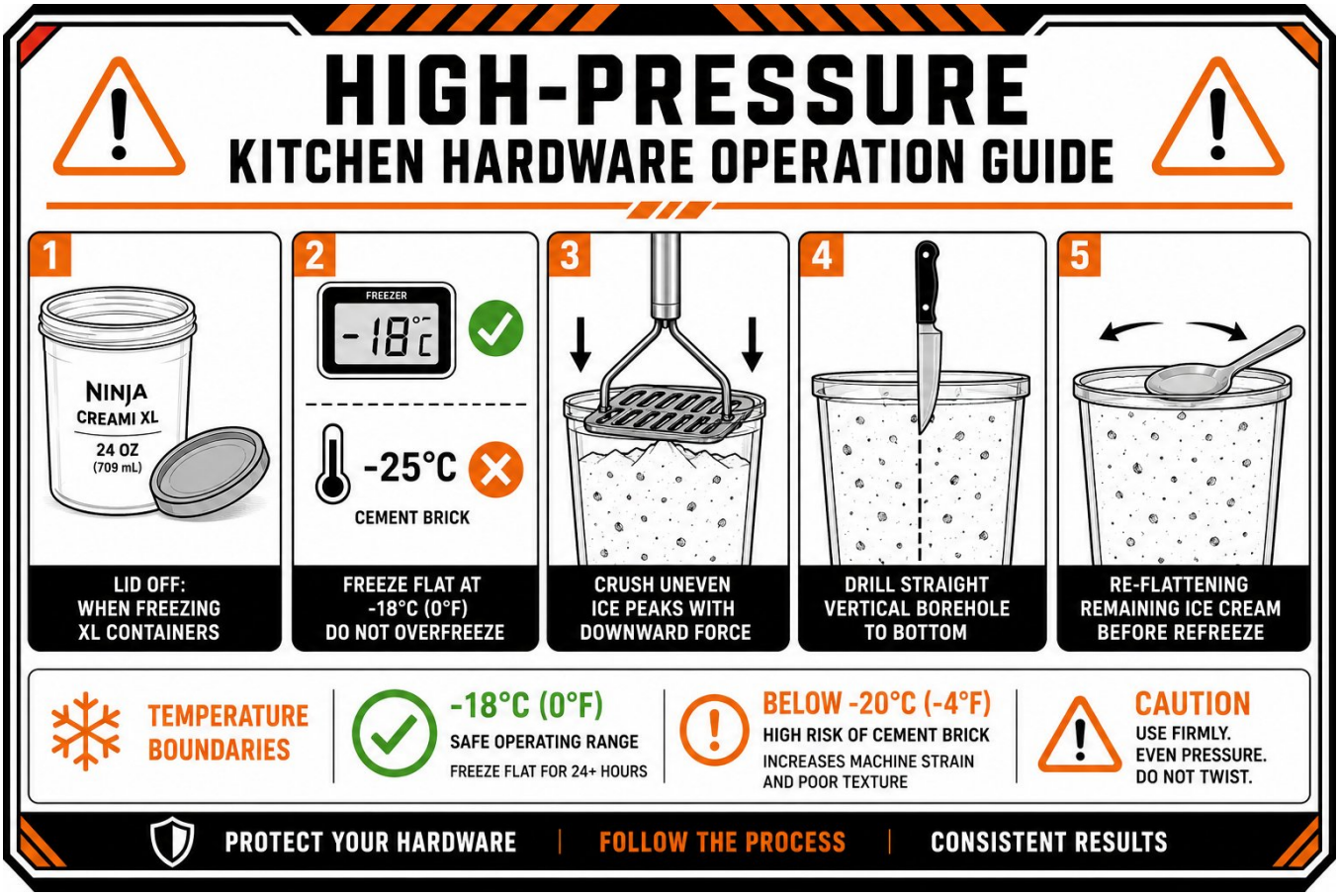
V194.1 — Silk base (archived ledger, 15 permutations)

#	Permutation	kcal	Protein	Fat	Net carbs
01–07	Hexapro fruit & classic line	464	79.5 g	11.0 g	13.0 g
04	Classic Vanilla Bean (Tier 1)	497	80.8 g	13.7 g	13.7 g
08–09	Midnight Mocha · Triple Mint Chip	434	82.5 g	9.5 g	7.0 g

10	Dark Chocolate Oreo	454	83.5 g	10.0 g	8.0 g
11	Peanut Butter Twist	584	92.5 g	19.0 g	10.0 g
12–13	Birthday Cake · Cinnamon Roll	554	75.0 g	17.0 g	26.5 g
14	Maple Walnut Cream	654	79.0 g	25.0 g	27.5 g
15	PB Fudge Crunch	614	91.5 g	21.0 g	17.0 g



KITCHEN CARD — The Strawberry Cream Profile, Silk-variant accounting as printed for the kitchen wall. Owner artwork.



KITCHEN CARD — The Velvet Banana Cream. Owner artwork.



THE RESULT — Strawberry Cream (Permutation 02), hard-scoop density off a dry re-spin. Owner photo.

CHAPTER 10 — VALIDATION REGISTER

Every load-bearing procedure in this system, checked against published references on 21 October 2026. Primary source: the NC501 Owner's Guide (SharkNinja). Supporting: hydrocolloid manufacturer monographs (Cape Crystal, JECFA, GELITA), sweetener technical literature (Tate & Lyle, Ingredion, Jungbunzlauer), Health Canada and USDA / FoodSafety.gov, and established community references (Tasty Travelers, Basics with Bails, Kroll's Korner, Maria Lucey RD). The Master Architecture's internal audit statuses agree with the external check at every point except emphasis — where they differ, this register governs the wording.

#	Claim in this system	Verdict	What the references say
1	Freeze ~24 h at −18 °C to −20 °C (0 °F to −4 °F); temper an over-cold pint	SUPPORTED	Manual: freeze "at least 24 hours"; freezer "between 9 °F and −7 °F" (−13 °C to −22 °C); chest freezers not recommended. For over-cold pints the manual prescribes a few minutes on the counter. The 30–60 s warm bath is community practice (Tasty Travelers, Maria Lucey). Note: −25 °C (−13 °F) is the EU manuals' limit; the US book's floor is −7 °F (−22 °C).
2	The centre hump must be levelled or it risks the drive	NO OFFICIAL SOURCE	The manual never mentions a hump. It does forbid processing a pint frozen at an angle or unevenly, and requires smoothing the surface. Hump mechanics come from community forensic practice; the most careful source (Tasty Travelers) distinguishes an angled freeze from a bump and notes bumps concentrate in low-fat/low-sugar bases — this system's bases.
3	Freeze lid-off for the first 2–4 hours	OWNER CANON	The manual instructs lid-on freezing. Lid-off is widely taught community technique for hump prevention. No Creami-specific food-safety ruling exists; general guidance favours covered storage for quality (freezer burn, odour). Retained as labelled owner canon — the single clear official-vs-system divergence in the book.
4	Lite Ice Cream / Sorbet for lean bases; never standard Ice Cream	SUPPORTED	Manual: Lite Ice Cream is for recipes low in sugar or fat or using sugar substitutes — keto and paleo named explicitly. Standard Ice Cream is for "traditionally indulgent" bases. The manual does

			not forbid lean bases on Ice Cream; community consensus is poor results, not damage.
5	Dry re-spin first; splash of 15–30 mL (0.5–1.0 fl oz) only if still powdery	**SUPPORTED**	The NC501 manual's re-spin procedure is dry — no liquid anywhere in it; re-spin is "often needed for very cold bases." Adding 1–2 tbsp (15–30 mL) appears in generic Ninja troubleshooting and universal community practice, sequenced after a dry attempt (Basics with Bails: add liquid only if still crumbly after two respins).
6	Scrape the icy wall ring after spin 1, then re-spin	**COMMUNITY**	Universally taught (Basics with Bails, Maria Lucey RD, Kroll's Korner) because the paddle misses the edges. The manual's only adjacent word is a caution against aggressive metal utensils in the pint — a gentle knife or spoon stays inside it.
7	Gum at 1.0 g (1000 mg) per ~650 g base; halve for blends/ pudding; dry-mix against clumps; overdose = slime	**SUPPORTED**	Published xanthan range in ice cream: 0.10–0.30% by weight; this system runs 0.15–0.23%. Overdose producing slimy/gummy texture is documented (Cape Crystal), and the Master Architecture's 0.25% w/w cross-link threshold in starch-bearing matrices is consistent with it. Dry pre-blending against "fish-eyes" is manufacturer guidance, near-verbatim.
8	Guar / tara interchangeable; pudding mix justifies the reduced gum dose	**SUPPORTED IN PRINCIPLE**	Guar: 0.1–0.3% typical in ice cream (JECFA); tara: 0.1–0.3%, "similar proportions to guar." Pudding mix is itself a stabilizer system; published base recipes instruct choosing one stabilizer, not stacking them. No source publishes a numeric reduction formula — the 1/8 tsp (0.5 g) figure is owner calibration, consistent with the principle.
9	Allulose (25–35 g · 0.9–1.2 oz per pint) keeps the pint softer	**SUPPORTED, NUANCED**	Tate & Lyle: allulose depresses the freezing point in frozen products; ~70% of sucrose's sweetness; browns readily. Ingredion: supports "smooth scoopability." The Master's 1.9× factor is the molecular-weight ratio. Nuance: erythritol's raw depression factor is higher (2.8 in some tables), but it crystallizes in

			storage and hardens pints (Jungbunzlauer); one industry test found allulose alone texturally poor, allulose + erythritol better. GI discomfort at high doses is documented.
10	Collagen / bone broth in the base	**REFINED**	Gelatin is a classic stabilizer; collagen peptides are not gelatin. GELITA's FAQ: hydrolyzed collagen cannot gel and its function is not texturizing. This book therefore classes broth + collagen (1 scoop · ~10 g each) as protein payload; the gums stabilize. Agrees with the Master Architecture's own REFINED status.
11	Pasteurized liquid egg whites, uncooked, 100 g (3.5 oz) per pint	**SUPPORTED**	Health Canada: use pasteurized egg products for foods that aren't heated. FoodSafety.gov: homemade ice cream is safe with pasteurized egg products. Liquid egg products are pasteurized by regulation. Keep refrigerated; respect dates; shell eggs are not pasteurized unless labelled.
12	Inclusions by borehole + Mix-In; delicate items hand-folded	**SUPPORTED**	Manual: Mix-In is designed for candies, cookies, nuts, cereal, frozen fruit; drill a 1½-inch (3.8 cm) hole to the bottom, ~1/4 cup (60 mL) maximum, chopped pieces; hard mix-ins stay intact — use minis or pre-chopped. "DO NOT process hard, loose ingredients or ice." This system's Tier 2/3 hand rules are stricter than the manual — texture doctrine, adopted from the Master Architecture.
13	Never exceed the Max Fill line; fill 600–645 g (21.2–22.8 oz)	**SUPPORTED**	Manual safety rule and prep steps both prohibit filling past Max Fill; overfilling is named the most typical cause of appliance overload in the motor-reset section. The gram window is the system's own precision layer on top of the line.
14	Cream of tartar, 1/8 tsp (~0.4 g), in fruit/bakery permutations	**FLAVOUR ONLY**	Its documented frozen-dessert role is acid-catalyzed sugar inversion — which requires ~20 minutes of boiling. Added cold, it inverts nothing and stabilizes nothing. Its verified function here is exactly what the recipes claim: acidity and

			brightness. Agrees with the Master Architecture’s REFINED status.
15	The warm base must taste over-sweet (taste-audit law)	**SUPPORTED**	Standard ice-cream doctrine in near-verbatim published form: cold dulls sweetness, so mixes “taste too sweet before freezing” and should be judged slightly over-sweet and over-flavoured before chilling (IceCreamCalc and standard troubleshooting literature).

What could not be verified

- Whether Ninja’s position on the centre hump has ever changed — successive manuals repeat identical uneven-freeze wording.
- A Creami-specific food-safety ruling on lid-off freezing (Claim 3 rests on general guidance plus community practice).
- A published numeric formula for the pudding-to-gum reduction (Claim 8).
- A validated home-scale ceiling for allulose per pint; reports of extreme doses failing to freeze are community anecdote.
- Live ninjakitchen.com FAQ text — the domain now redirects to sharkninja.com; the NC501 Owner’s Guide was read in full instead.

CHAPTER 11 — RECONCILIATION REGISTER

Four source editions feed this book. The Culinary Engine V198.4 governs formulation; the Master Architecture V186.00 governs the food-science and engineering layer; V194.1 contributes the physics, the scaling protocol and the Silk variant; the V1.0 consolidated edition contributes the external validation. Rulings:

Item	Earlier / alternate	Governing	Ruling
Base milk	Silk Protein Almond & Cashew (V194.1)	Kirkland Ultra-Filtered 2% (V198.4)	Kirkland governs; Silk preserved as certified variant (Ch. 4) with its own ledger (Ch. 9).
Stabilizer dose	Fixed 1/4 tsp xanthan (V194.1)	Calibrated 1/4 tsp isolate / 1/8 tsp blend-pudding (V198.4)	Calibration governs; Master's 0.25% w/w slime threshold (Ch. 2) is its physical basis.
Gum true measure	1.0–1.5 g / 0.5–0.7 g ranges (V198.4 ops text)	1.0 g / 0.5 g exact (Master Architecture)	Exact figures govern the dual-measure tables; the scale-and-level protocol (Ch. 5, Step 1) is what makes exactness achievable.
Fill specification	"Below Max Fill" only (V194.1 / V198.4)	600–645 g (21.2–22.8 oz) window (Master; matches owner blueprint plates)	Gram window adopted as the system spec (Ch. 4, Step 3 verification).
Freeze structure	Lid-off 2–4 h + lid to full 24 h (V194.1)	Lid-off 2–4 h + 16–24 h total (V198.4)	Printed as 24 h total standard, 16 h absolute minimum — the conservative reading of both, and Ninja's own number.
Warm bath	Exactly 30 s (V194.1)	30–60 s (V198.4)	30–60 s window governs; 30 s default.
Re-spin liquid	Silk milk splash (V194.1)	Kirkland milk splash (V198.4)	Splash follows the base in use; dry re-spin precedes either (Ch. 10, Claim 5).
Inclusions	Borehole + hand-fold notes (V198.4)	Three-tier architecture (Master)	Three-tier system governs (Ch. 7); it formalizes, not contradicts, the earlier notes.
Permutation count & names	15 (V194.1)	18 (V198.4); Master prefixes 16–18 "SF"	18 govern; printed under their V198.4 "Zero-Sugar" names, which the kitchen cards use.

Collagen & cream of tartar	Implied structural roles (early drafts)	REFINED: protein payload; flavour acidifier (Master + external check)	Both editions agree after refinement; Ch. 2 and Ch. 10 word them identically.
L-Glutamine	Removed from base; separate beverage (V194.1)	Absent (V198.4)	Consistent: not a base component in any governing edition.

CHAPTER 12 — HARDWARE SAFETY & THE HALL OF MISTAKES

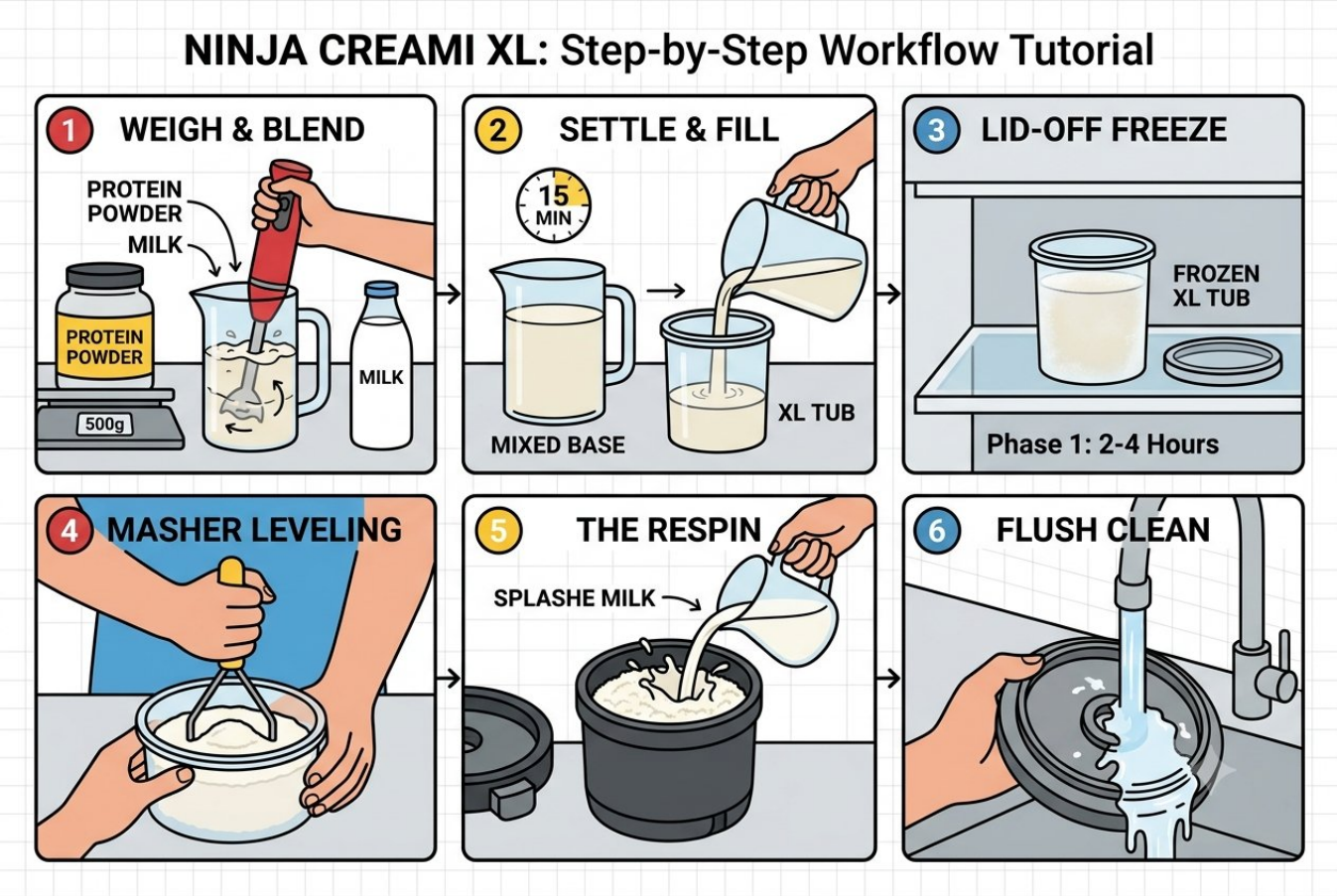
Critical limits

- **Thermal window:** process pints frozen at −18 °C to −20 °C (0 °F to −4 °F). Below −22 °C (−8 °F), temper before spinning; below −25 °C (−13 °F) the base is a cement brick.
- **Fill window:** 600–645 g (21.2–22.8 oz), never past Max Fill (Ch. 10, Claim 13).
- **Surface law:** the block's top must be level before the tub enters the machine.
- **Program law:** Lite Ice Cream or Sorbet only, for every base in this system.
- **Sanitation law:** hot-water spindle flush immediately after every spin — protein mist in the drive recess becomes biofilm and lock failure.

The Hall of Mistakes — forensic register

#	Documented error	Root cause	Override protocol
01	The Stabilizer Slime Trap	Gum past 1/8 tsp (0.5 g) on casein blends or pudding bases cross-links past the 0.25% w/w threshold.	Enforce the 1/8 tsp ceiling; level with the flat bar; weigh it.
02	Hidden Sugar / Agave Trap	Vanilla pastes carried in agave or glucose syrup quietly break zero-sugar.	Audit the carrier: glycerin or alcohol only.
03	Soft Inclusion Obliteration	Machine Mix-In pulverizes dough bits and bar chunks to mud.	Tier 2 hand-fold, spoon, post-spin (Ch. 7).
04	The Premature Spin	Core under 16 h or not 100% solid: the shaving profile never engages, texture collapses to slush.	Solid-block check is mandatory; 24 h standard.
05	Excessive Thaw Bath	Past 60 s the perimeter liquefies while the core stays rock-hard.	30–60 s window; edges must stay structurally firm.
06	One-Program Bias	Standard Ice Cream is calibrated for soft, fat-rich bases; lean blocks need the higher-shear programs.	Lite Ice Cream or Sorbet, always (Claim 4).
07	Unnecessary Liquid Splash	Milk dumped into a powdery first spin before friction has had its chance turns the pint to soup.	Dry re-spin first; splash 15–30 mL (1–2 tbsp) only if still powdery.
08	Direct Powder Dumping	Gum or protein into cold liquid forms hydrophobic glue pockets.	Dry-whisk first; stream into the vortex over 90 seconds.
09	The Centre Hump Violation	A convex dome deflects the descending shaft and loads the drive laterally.	Shatter and level with the Deiss masher — straight down, never twist.
10	Overflow Freezing	Past Max Fill, expansion chokes the motor and kills overrun.	Weigh the fill: 600–645 g (21.2–22.8 oz), every pint.

11	The Perimeter Ring Omission	The paddle cannot reach the ~1 mm (0.04 in) frost ring on the wall; unmixed, it reads as ice flakes in every spoonful.	Gentle knife scrape after spin 1; fold into the core; re-spin.
12	Spindle Cleanliness Failure	Dairy/protein residue dries inside the lid’s locking cavity; springs seize, biofilm grows.	Hot flush through the spindle aperture immediately post-spin. Critical, not optional.



WALL POSTER — High-Pressure Kitchen Hardware Operation Guide: lid-off freezing, the −18 °C (0 °F) boundary, masher levelling, borehole, re-flattening. Owner artwork.

APPENDIX A — EQUIPMENT & PANTRY

Laboratory tooling manifest

- **Ninja Creami XL Deluxe (NC501 series)**, 800 W · 24 oz (709 mL) Deluxe Pints & lids
- **Taylor digital kitchen scale** — 2000 g (4.4 lb) capacity, 0.1 g (0.004 oz) readability
- **Magnetic measuring spoons (1/8 and 1/4 tsp) + flat metal levelling bar** — the calibration instruments
- **KitchenAid variable-speed hand blender** + 1000 mL (33.8 fl oz) glass mixing jug
- **Stainless steel mixing bowl, 2.5 L (84.5 fl oz)** — dry matrix
- **Deiss heavy-duty stainless steel potato masher** — flat-plate levelling
- **68 oz (2.0 L) Le'raze glass pitcher** — 3× batch scaling
- **VOLTRX electric shaker** — prep-station support

The emulsion library (as shelved)

- LorAnn Bakery Emulsions: Coffee (118 mL / 4 fl oz), Vanilla, Lemon, Natural Peppermint, Strawberry, Butter Vanilla, Blueberry (60 mL / 2 fl oz each)
- Vanillux Vanilla Paste (70 g / 60 mL)
- Stabilizers in rotation: Bob's Red Mill xanthan gum; tara; guar
- OrgunFun pasteurized egg white protein · Chef's Choice Himalayan pink rock salt · Mio liquid enhancer

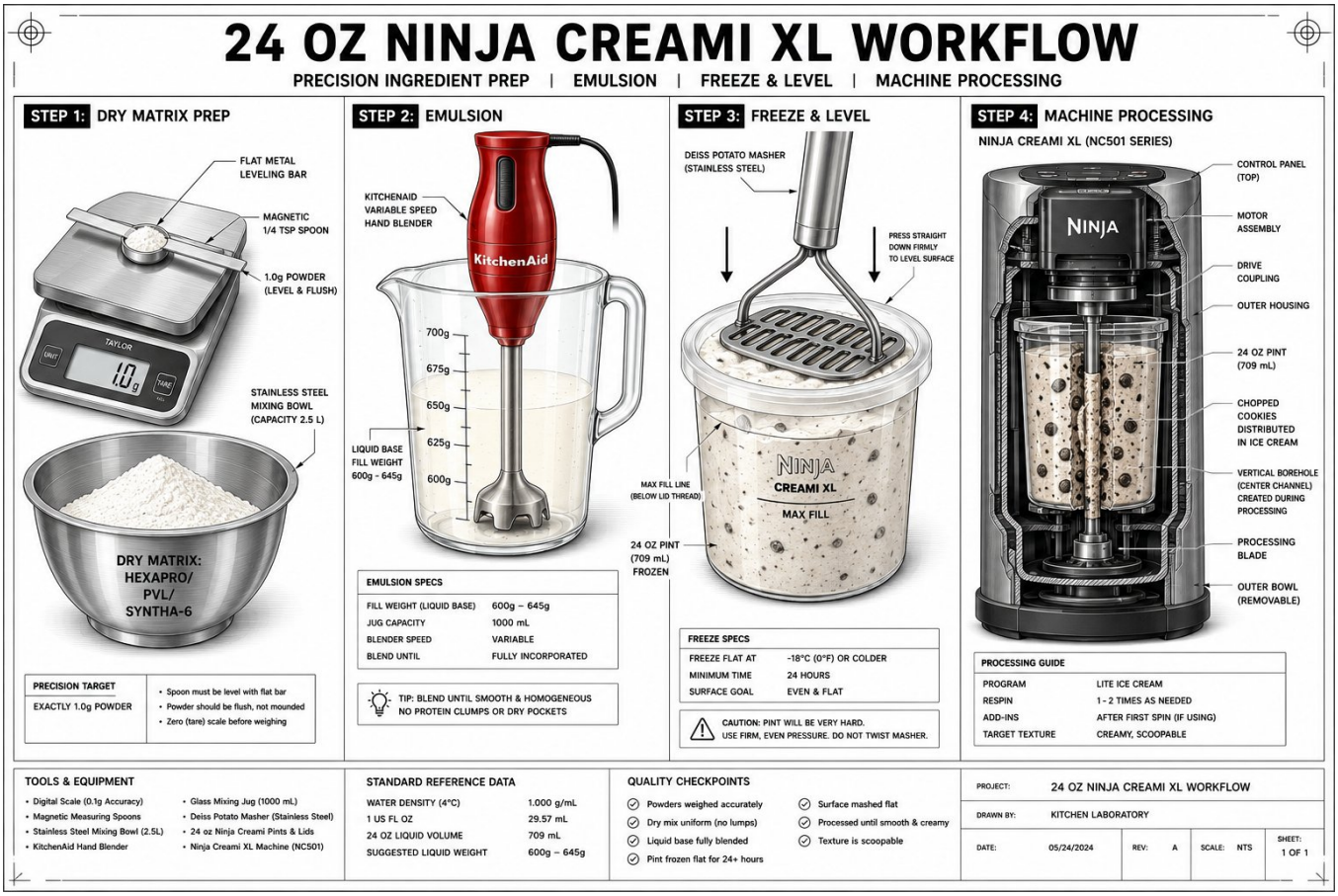


PLATE — 24 oz Ninja Creami XL Workflow, engineering sheet: dry matrix prep · emulsion · freeze & level · machine processing (NC501 cross-section). Owner artwork.

THE STRAWBERRY CREAM PROFILE

BASE INTEGRATION:

150ml Boiling Water
1 Scoop Bone Broth
1 Scoop Marine Collagen
200ml Silk Protein Milk
100g Liquid Egg Whites



MATRIX ADD-INS:

1.5 Scoops Hexapro Vanilla
3-4 tbsp Sweetener
1/4 tsp Xanthan Gum
1 Pinch Salt
1 tsp LorAnn Strawberry Emulsion
1/2 tsp Vanillux Paste

TOTAL PINT MACROS:

464 kcal | 79.5g Protein | 11g Fat Fat | 13g Net Carbs

PROCESSING:

1. Dissolve broth and collagen in boiling water.
2. Blend cold liquids and dry matrix matrix 90s.
3. Settle 15m. 4. Freeze uncovered 4h, then lock lid at -18C fo 24h. Excouverer. 5. Flatten surface with Deiss masher.
6. Spin Lite Cream, 6. Spin Lite Ice Cream, execute Re-Spin.

PLATE — Processing workflow schematic, grid layout: dry prep → liquid mix → freeze & level → processing. Owner artwork.

APPENDIX B — REFERENCE SOURCES

Validation pass, 21 October 2026. Sources listed against the Chapter 10 claims they support.

- **NC501 Owner's Guide** (SharkNinja) — freeze duration and temperature range, lid-on instruction, level-surface rule, Lite Ice Cream definition, dry re-spin procedure, mix-in procedure and limits, Max Fill rules, overload protection, counter tempering. Claims 1–6, 12, 13.
- **Cape Crystal Brands** — xanthan conversion and troubleshooting: ice-cream usage 0.10–0.30%, slime at overdose, fish-eye prevention by dry pre-mixing; tara gum proportions. Claims 7, 8.
- **JECFA monograph; Altrafine; Sosa** — guar gum in ice cream, 0.1–0.5%. Claim 8.
- **Tate & Lyle; Ingredion** — allulose freezing-point depression, ~70% sweetness, browning, scoopability. **Jungbunzlauer** — erythritol crystallization vs allulose in light protein ice cream. Claim 9.
- **GELITA / gelatininfo.com** — gelatin as classic stabilizer; collagen peptides cannot gel and are not texturizers. Claim 10.
- **Health Canada; USDA / FoodSafety.gov** — pasteurized egg products for unheated foods and homemade ice cream. Claim 11.
- **Tasty Travelers; Basics with Bails; Maria Lucey RD; Kroll's Korner; George Eats** — lid-off freezing, hump analysis, tempering, re-spin sequencing, wall scrape, stabilizer choice. Claims 1, 2, 3, 5, 6, 8.
- **IceCreamCalc knowledge base** — cold dulls sweetness; the over-sweet unfrozen base. Claim 15.
- **Pastry Sampler; EatSip365** — cream of tartar as heated sugar-inversion catalyst. Claim 14.

COLOPHON

Ninja Creami XL Protein Ice Cream Guidebook — **Final Edition, Version 2.0 · 21 October 2026**. PROJECT 100 Kitchen Laboratory. Synergized from: Master Operations Manual / Culinary Engine V198.4 · Master Architecture V186.00 · Operations Manual V194.1 · Consolidated Edition V1.0 with external validation. Calibrated for Bogdan Dospinescu — 2,780 kcal ceiling · 252 g protein ceiling · 69 g lipid ceiling · terminal target 100.0 lb SMM. Companion volumes: PROJECT 100 Encyclopedia V3.4 · Guidebook V3.5. Standing rule of the house: overview and verification over copy-paste — and every disagreement printed in the open.